

The EEG display consists of an arrangement of channels.

Each channel is made up of two individual recording electrodes that are glued to the patient's scalp and record the electrical activity of the underlying cortex. The EEG machine constantly calculates the difference in electrical potential between these two electrodes, amplifies and then displays it as a deflection (wave) on the screen.

For example, if E1 and E2 are two scalp electrodes that make up channel E1-E2, then the direction and amplitude of the wave displayed by channel E1-E2 depends on the relative difference in electrical potentials at E1 and E2 (E1-E2).

Conventionally,

1. if the potential at E1 is less than E2 (E1 is relatively negative) then their difference (E1-E2) will be negative – manifesting as an upward deflection;
2. if the potential at E2 is less than E1 (E2 relatively negative) then their difference (E1-E2) will be positive – manifesting as a downward deflection;
3. if both E1 and E2 are equipotential or inactive ($E1 = E2$) then their difference (E1-E2) will be neutral – manifesting as no deflection.

Therefore, the display pointer always deflects toward the relatively smaller (more negative/less positive) electrode. If this electrode is E1, the deflection is upward (negative) and if it is E2, the deflection is downward (positive). If both electrode potentials are equal or inactive, there is no deflection [1]. Figure 2.1 illustrates polarity.

Chapter Summary

1. The polarity (direction and amplitude of deflection) depends on the relative difference between the two electrode potentials.
2. The pointer always deflects to the electrode with the relatively smaller potential (more negative/less positive).

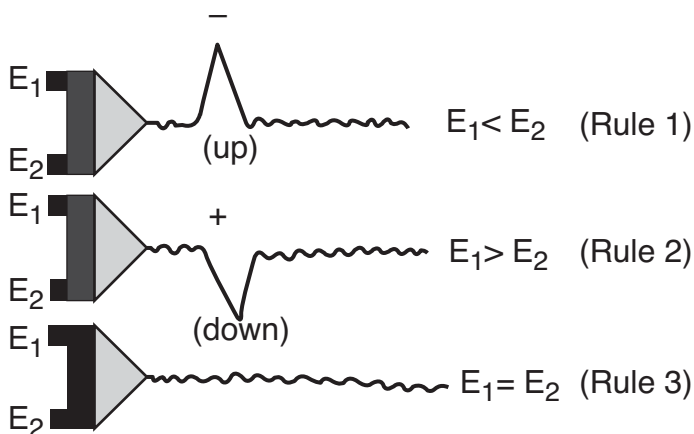


Figure 2.1 Upward (negative), downward (positive), and no deflection (neutral).

3. Upward deflection is surface negative, and downward deflection is surface positive.

Reference

1. Niedermeyer E. The EEG signal: polarity and field determination. In *Electroencephalography: basic principles, clinical applications, and related fields* (pp. 79–83). Oxford University Press, Oxford; 1987.